Fractions

Curriculum Ready

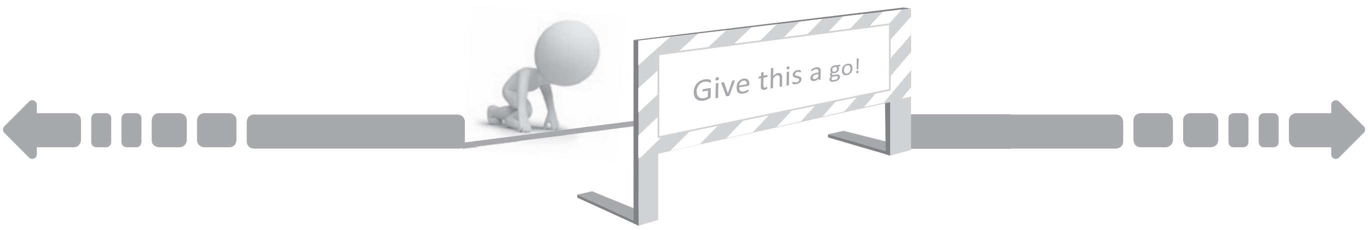
ACMNA: 152, 153, 154, 155

Mathletics

www.mathletics.com

Fractions allow us to split things into smaller equal sized amounts.

Write down two occasions where you have had to split something up evenly between family members or friends. Describe how you made sure this was done fairly each time.



Q For one particular school:
There are 256 students in Year 7.
The Year 8, 9 and 10 groups all have half the number of students than the year just below them.
How many students are there at this school in Years 7 to 10?

Work through the book for a great way to do this

Proper fractions

Proper fractions represent parts of a whole number or object.

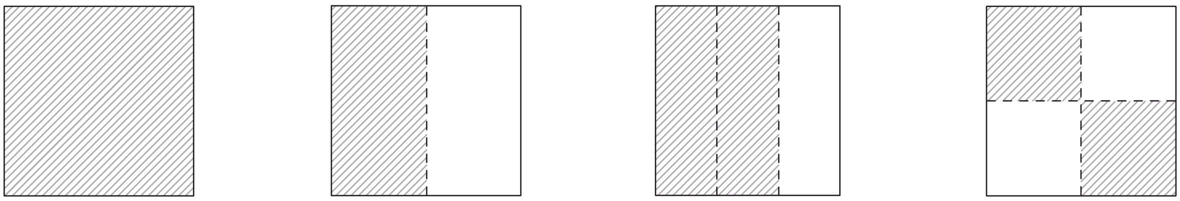
numerator → $\frac{1}{2}$ ← number of equal parts **you have**

denominator → ← **total** number of equal parts

The numerator is always smaller than or equal to the denominator in proper fractions.

Let's look at some equally sized shaded shapes.

(i) Write a fraction for the shaded parts of the squares below:



1 whole square | Split into 2 equal parts | Split into 3 equal parts | Split into 4 equal parts

The number of equal parts shaded

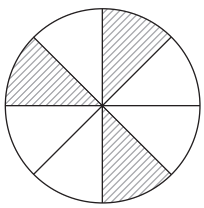
$\frac{1}{1} = 1$ $\qquad \frac{1}{2}$ $\qquad \frac{2}{3}$ $\qquad \frac{2}{4}$

The total number of equal parts

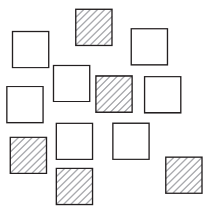
Larger denominator $=$ smaller equal parts

(ii) Shade these to match the fraction:

$\frac{3}{8}$

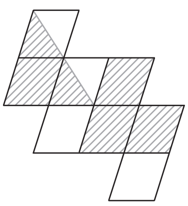


$\frac{5}{12}$ ← Shaded parts
← Total equal parts



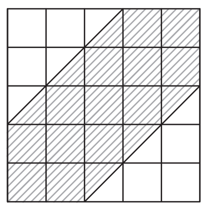
(iii) Include at least two half-shapes when shading these to match the fraction:

$\frac{5}{8}$



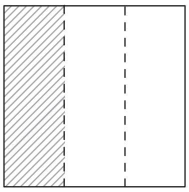
Two halves $= 1$ whole

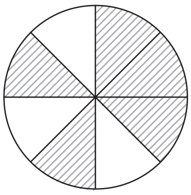
$\frac{16}{25}$

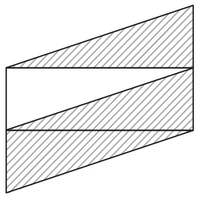


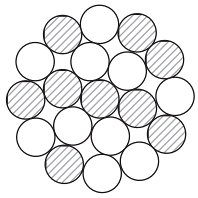
Proper fractions

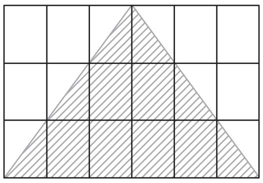
1 What fraction of these equal-sized shapes have been shaded?

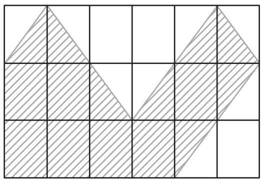
a 

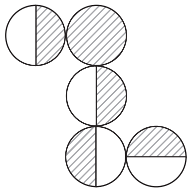
b 

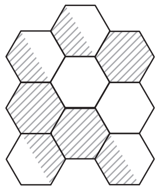
c 

d 

e 

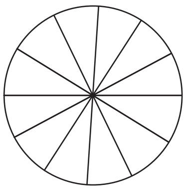
f 

g 

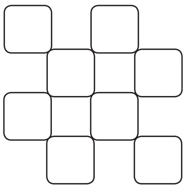
h 

2 Shade these to match the given fraction:

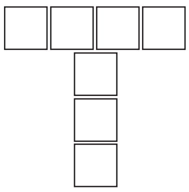
a $\frac{5}{12}$



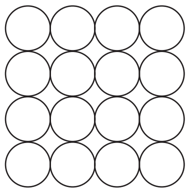
b $\frac{8}{8}$



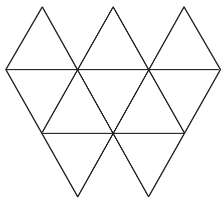
c $\frac{3}{7}$



d $\frac{11}{16}$

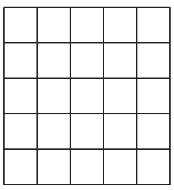


e $\frac{4}{10}$

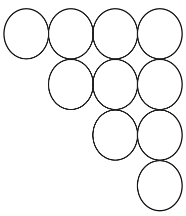


3 Shade these to match the given fraction, including **at least** one pair of half-shapes:

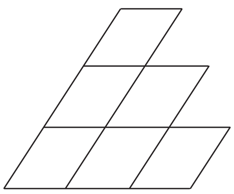
a $\frac{9}{25}$



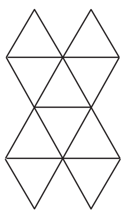
b $\frac{3}{10}$



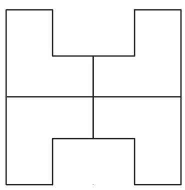
c $\frac{5}{6}$



d $\frac{7}{10}$



e $\frac{1}{4}$



Proper fractions

4. Draw and shade diagrams with equal sized shapes to represent each of these fractions:
   (i) Shading whole shapes only.
   (ii) Including at least one pair of half-shaded shapes.

a. $\frac{3}{5}$

(i)

(ii)

b. $\frac{2}{9}$

(i)

(ii)

c. $\frac{5}{8}$

(i)

(ii)

d. $\frac{4}{7}$

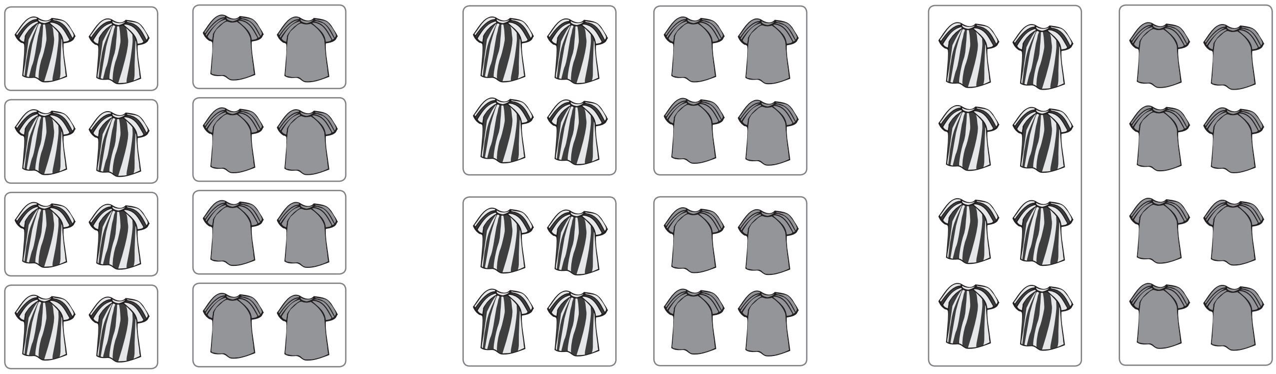
(i)

(ii)

Equivalent proper fractions

These are fractions with different numbers that represent the same amount.
For example, two fitness teams do three sessions of training in the same park.

Session 1: Grouped in pairs **Session 2:** In groups of four **Session 3:** Grouped as a whole team



$\frac{4}{8}$ $\frac{2}{4}$ $\frac{1}{2}$

Fraction of training groups wearing striped (or plain) shirts in each session.

The **groups change size** but the total **number of people** training **remains the same**

$\therefore \frac{4}{8} = \frac{2}{4} = \frac{1}{2} =$ Equivalent fractions

We find equivalent fractions by dividing/multiplying the numerator and denominator by the same number.

Write an equivalent fraction for each of these using the multiplication or division given in square brackets

(i) $\frac{3}{5}$ $[\times 3]$

$\frac{3\times 3}{5\times 3} = \frac{9}{15}$

$\therefore \frac{3}{5}$ and $\frac{9}{15}$ = equivalent fractions

(ii) $\frac{12}{32}$ $[\div 4]$

$\frac{12\div 4}{32\div 4} = \frac{3}{8}$

$\therefore \frac{12}{32}$ and $\frac{3}{8}$ = equivalent fractions

Simplify these fractions by dividing the numerator and denominator by the highest common factor (HCF)

Simplify = Find the smallest equivalent fraction. ☺

HCF: the largest number that divides into both exactly

(i) $\frac{3}{9}$

HCF for 3 and 9 is: 3

$\therefore \frac{3\div 3}{9\div 3} = \frac{1}{3}$

$\therefore \frac{1}{3}$ is the simplest equivalent fraction to $\frac{3}{9}$

(ii) $\frac{18}{24}$

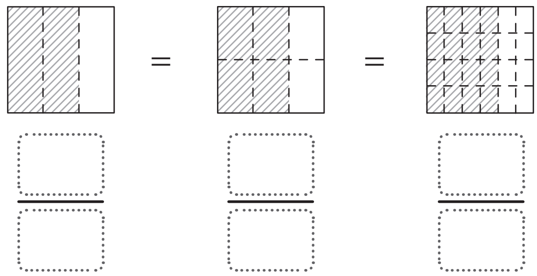
HCF for 18 and 24 is: 6

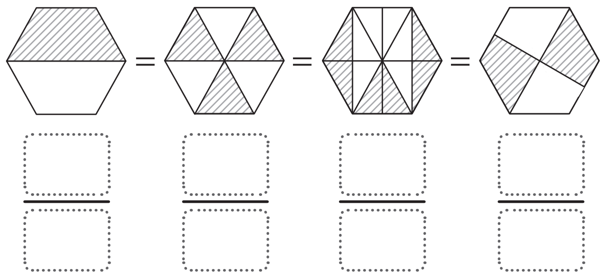
$\therefore \frac{18\div 6}{24\div 6} = \frac{3}{4}$

$\therefore \frac{3}{4}$ is the simplest equivalent fraction to $\frac{18}{24}$

Equivalent proper fractions

1. Write the equivalent fractions represented by these equally-sized shaded areas:

a 

b 

2. Write an equivalent fraction for each of these using the multiplication or division given in square brackets:

a $\frac{1}{4}$ $[\times 5]$

b $\frac{8}{10}$ $[\div 2]$

c $\frac{3}{5}$ $[\times 3]$

d $\frac{12}{24}$ $[\div 6]$

3. Simplify these fractions by dividing the numerator and denominator by the highest common factor (HCF).

a $\frac{16}{20}$

b $\frac{8}{32}$

4. Simplify these two fractions.

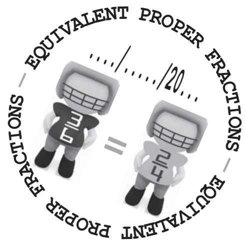
a $\frac{14}{21}$

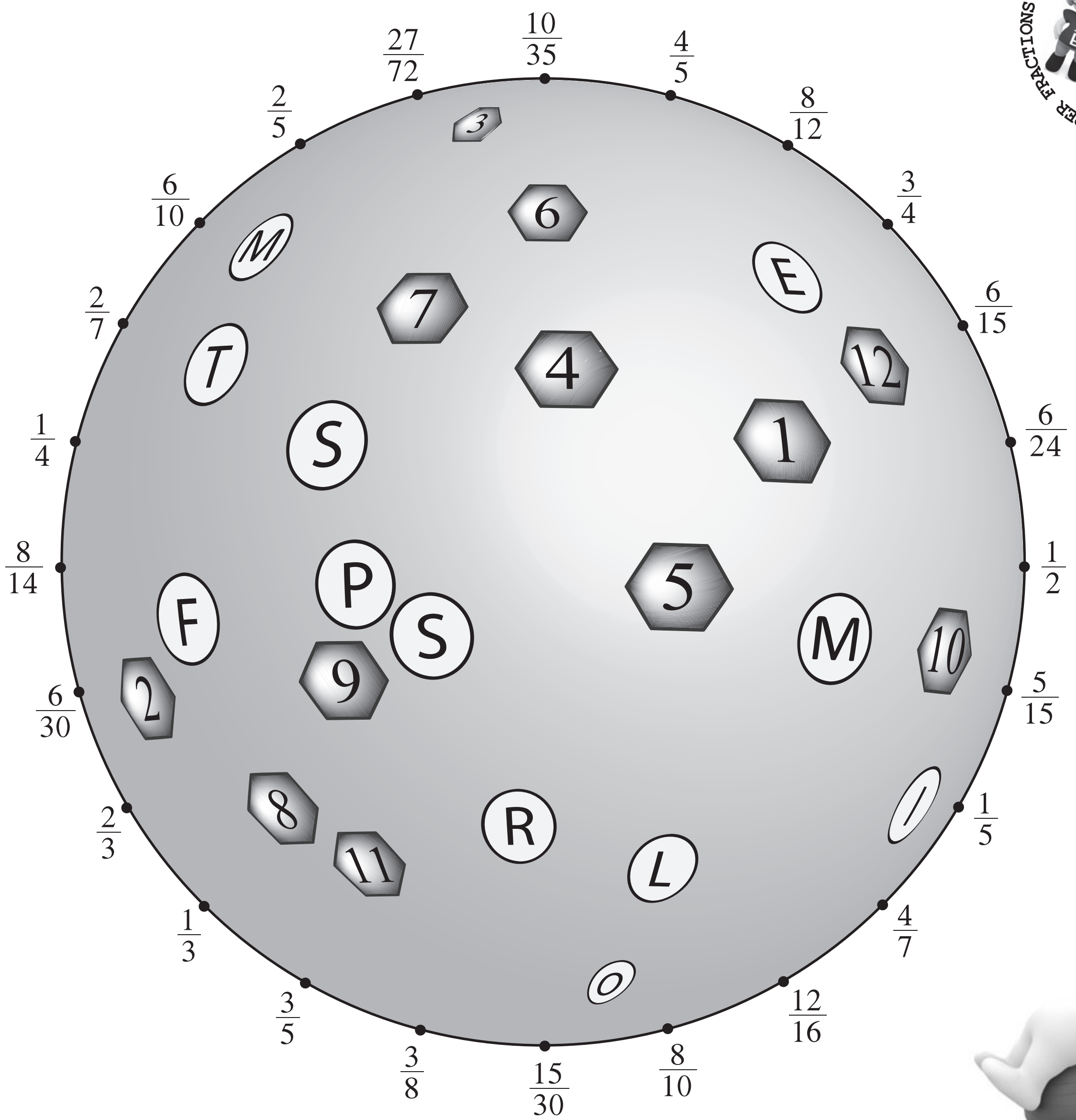
b $\frac{16}{24}$

5. Are the fractions $\frac{14}{21}$ and $\frac{16}{24}$ from question 4 equivalent fractions? Briefly explain your answer.

Equivalent proper fractions

6. Match the pair of equivalent fractions below by joining them with a straight line.
Solve the puzzle by matching the letter with the number each straight line passes through.





1 2 3 4 5 6 7 8   9 10 11 12

Improper fractions and mixed numerals

An improper fraction has a bigger numerator (top) than denominator (bottom)

$$\frac{3}{2} \longleftarrow \text{Improper fractions} \longrightarrow \frac{5}{4}$$

numerator $>$ denominator

$>$ means 'bigger than'

Mixed numerals have a whole number and a proper fraction.

$$1\frac{1}{2} \longleftarrow \text{Mixed numerals} \longrightarrow 1\frac{1}{4}$$

A 'mix' of whole numbers and proper fractions.

Mixed numerals are simplified improper fractions.

Simplify these

Improper fractions to mixed numerals

(i) $\frac{5}{3}$

$$\frac{5}{3} = 5 \div 3$$

$\frac{\text{numerator}}{\text{denominator}}$ = numerator $\div$ denominator

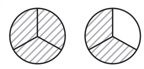
$$= 1\,r\,2$$

$$= 1\frac{2}{3}$$

remainder

Whole number answer

same denominator



picture form

(ii) $\frac{14}{4}$

$$\frac{14}{4} = \frac{7}{2} = 7 \div 2$$

Simplify if possible

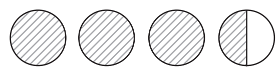
$$= 3\,r\,1$$

$$= 3\frac{1}{2}$$

remainder

Whole number answer

same simplified denominator



Mixed numerals to improper fractions

(i) $1\frac{2}{3}$

$$1\frac{2}{3} = \frac{3 \times 1 + 2}{3}$$

$$= \frac{5}{3}$$

same denominator

(ii) $2\frac{1}{5}$

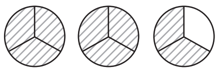
$$2\frac{1}{5} = \frac{5 \times 2 + 1}{5}$$

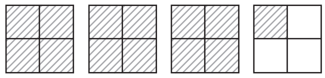
$$= \frac{11}{5}$$

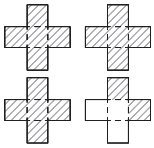
same denominator

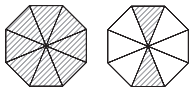
Improper fractions and mixed numerals

1. Write the mixed numerals represented by these shaded diagrams:

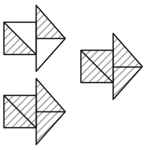
a  =

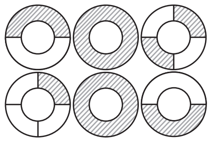
b  =

c  =

d  =

Make sure you write the fraction in simplest form where possible.

e  =

f  =

2. Simplify these improper fractions by writing them as mixed numerals.

a $\frac{12}{5}$ b $\frac{14}{3}$ c $\frac{23}{2}$

3. Write these fractions in simplest form first, then change to the mixed numerals.

a $\frac{15}{9}$ b $\frac{21}{14}$ c $\frac{18}{16}$

4. Write the equivalent improper fraction for these mixed numerals.

a $1\frac{1}{2}$ b $2\frac{3}{4}$ c $4\frac{4}{5}$

5. Write the equivalent improper fraction for these mixed numerals after first simplifying the fraction parts.

a $4\frac{2}{12}$ b $2\frac{6}{24}$ c $25\frac{24}{72}$

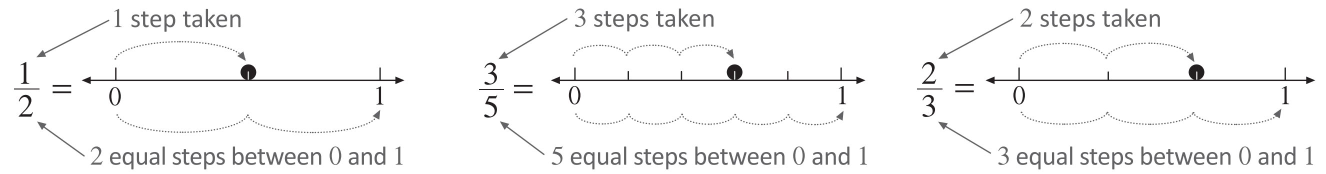
Fractions on the number line

Proper fractions represent values between 0 and 1 on a number line.

$$\frac{1}{2}$$
1 ← number of equal steps taken between 0 and 1
2 ← **total** number of **equal** steps between 0 and 1

Mark equal-sized steps matching the denominator between 0 and 1, then plot the fraction using the numerator.

Display the fractions $\frac{1}{2}$, $\frac{3}{5}$ and $\frac{2}{3}$ on these number lines

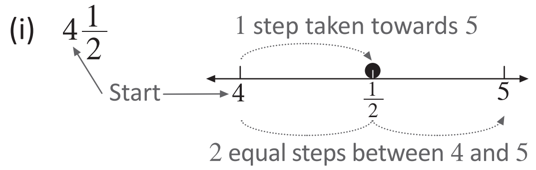


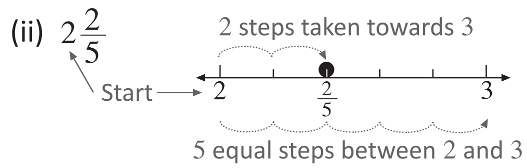
For mixed numerals, plot the fraction between the given whole number and the next whole number.

Start from this whole number → $3\frac{1}{2}$
1 ← number of equal steps towards the next whole number '4'
2 ← **total** number of equal steps between '3' and the next whole number '4'

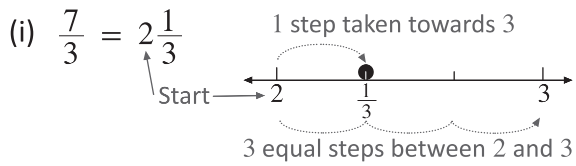
Display and read these fractions on a number line

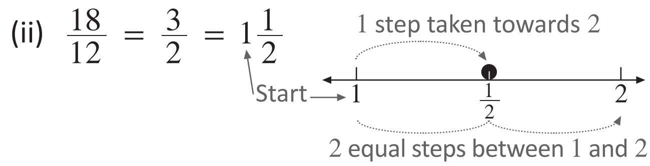
Mixed numerals



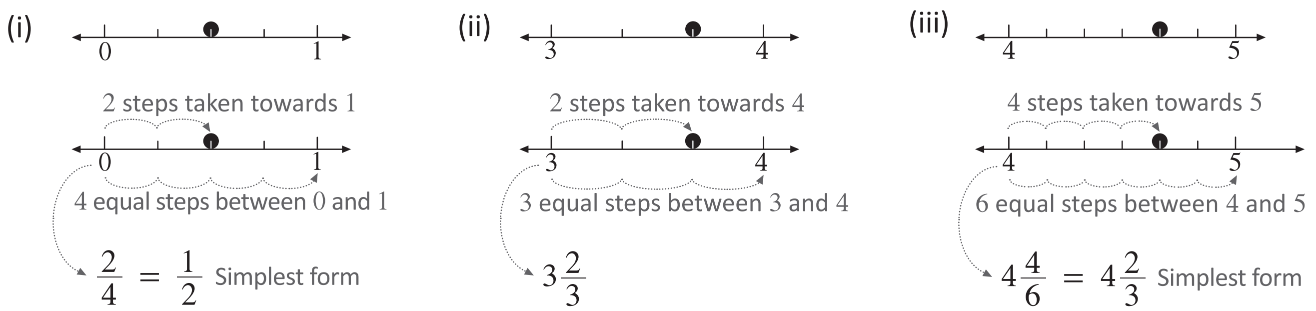


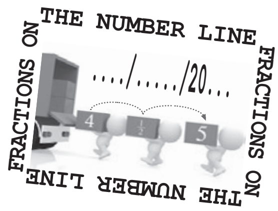
Improper fractions – simply change to the equivalent mixed numeral first then show on the number line





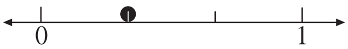
Write down the fraction displayed on these number lines

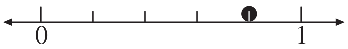


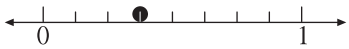


Fractions on the number line

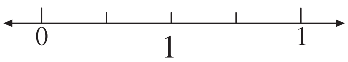
1 What proper fraction do the following points on the number line represent?

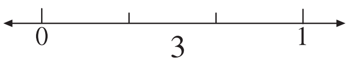
a 

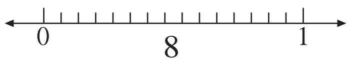
b 

c 

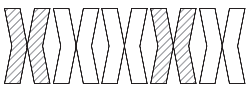
2 Display these fractions on a number line:

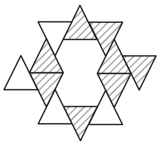
a 
$\frac{1}{4}$

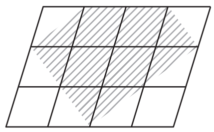
b 
$\frac{3}{3}$

c 
$\frac{8}{15}$

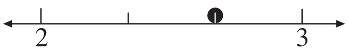
3 Write and display the fraction of equal shapes shaded on a number line for these diagrams:

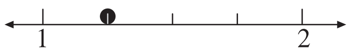
a 


b 


c 


4 Write the mixed numeral and equivalent improper fraction for the dots plotted on these number lines:

a 

b 

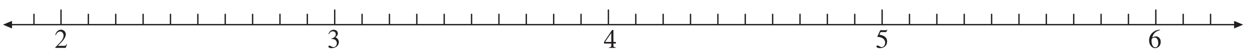
c 

5 Display these improper fractions on the number line:

a $\frac{27}{10} =$

b $\frac{11}{2} =$

c $\frac{22}{5} =$

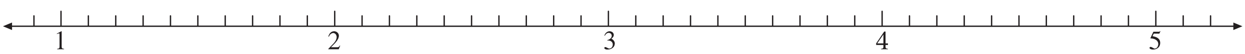


6 Display these on the number line after changing to equivalent fractions in simplest form first.

a $\frac{42}{15} = \quad = $

b $\frac{63}{18} = \quad = $

c $\frac{110}{25} = \quad = $



Reciprocal fractions

Original fraction $\longrightarrow \frac{2}{5}$ swap $\longrightarrow \frac{5}{2} \longleftarrow$ Reciprocal fraction

Write the reciprocal of these fractions

(i) $\frac{3}{4}$

$\frac{3}{4} \rightarrow \frac{3}{4}$ swap $\rightarrow \frac{4}{3} \longleftarrow$ Reciprocal fraction

(ii) $\frac{18}{8}$

$\frac{18}{8} = \underbrace{\frac{9}{4}}_{\text{Simplified}} \rightarrow \frac{9}{4}$ swap $\rightarrow \frac{4}{9} \longleftarrow$ Reciprocal fraction

For mixed numerals, change to an improper fraction first then write the reciprocal.

Mixed numeral $\longrightarrow 3\frac{1}{2} \longrightarrow \underbrace{\frac{7}{2}}_{\text{Improper fraction}} \longrightarrow \frac{7}{2}$ swap $\longrightarrow \frac{2}{7} \longleftarrow$ Reciprocal fraction

Whole/mixed number examples: Always write as a fraction first.

Write the reciprocal of these

(i) 3

$3 = \underbrace{\frac{3}{1}}_{\text{Whole number as a fraction}} \rightarrow \frac{3}{1}$ swap $\rightarrow \frac{1}{3} \longleftarrow$ Reciprocal fraction

(ii) $2\frac{3}{4}$

$2\frac{3}{4} \rightarrow \underbrace{\frac{11}{4}}_{\text{Improper fraction}} \rightarrow \frac{11}{4}$ swap $\rightarrow \frac{4}{11} \longleftarrow$ Reciprocal fraction

(iii) $4\frac{9}{15}$

$4\frac{9}{15} \rightarrow \underbrace{4\frac{3}{5}}_{\text{Simplified fraction}} \rightarrow \underbrace{\frac{23}{5}}_{\text{Improper fraction}} \rightarrow \frac{23}{5}$ swap $\rightarrow \frac{5}{23} \longleftarrow$ Reciprocal fraction

We will see why we find the reciprocal a little later on in this booklet.

It is used when dividing an amount by a fraction.

Reciprocal fractions

1. Write the reciprocal for these fractions:

   a $\frac{2}{3}$ b $\frac{11}{7}$ c $\frac{6}{19}$ d $\frac{15}{4}$

2. Write the reciprocal, then simplify these fractions:

   a $\frac{6}{10}$ b $\frac{14}{8}$ c $\frac{12}{18}$ d $\frac{25}{10}$

3. Write the reciprocal of these:

   a $\frac{1}{5}$ b $\frac{1}{9}$ c 2 d 4

4. Write the reciprocal of these mixed numerals:

   a $2\frac{1}{3}$ b $3\frac{2}{5}$ c $1\frac{5}{9}$

5. Write the reciprocal of these mixed numerals after first writing as a fraction:

   a $3\frac{2}{10}$ b $1\frac{10}{12}$ c $2\frac{9}{21}$

6. Write the reciprocal of these fractions as a simplified mixed numeral:

   a $\frac{10}{48}$ b $\frac{12}{66}$ c $\frac{15}{115}$

Comparing fractions

This is where we see which fractions are larger than others.

$$\frac{1}{2} \quad \text{or} \quad \frac{1}{3}$$

Write equivalent fractions by changing the denominators to their LCM.

Lowest Common Multiple (LCM)
The smallest value that appears in both times tables

$$\frac{1\times 3}{2\times 3} \longrightarrow \frac{3}{6} \quad \text{or} \quad \frac{2}{6} \longleftarrow \frac{1\times 2}{3\times 2}$$

Since they have the same denominator, now just compare their numerators.

bigger > smaller

$$\frac{3}{6} > \frac{2}{6}$$

$$\therefore \frac{1}{2} > \frac{1}{3}$$

Compare the size of these fractions

(i) $\frac{2}{3}$, $\frac{3}{4}$ and $\frac{5}{12}$

$$\frac{2}{3} \quad , \quad \frac{3}{4} \quad \text{and} \quad \frac{5}{12}$$

LCM of denominators $= 12$

$$\frac{2\times 4}{3\times 4} \longrightarrow \frac{8}{12} \quad , \quad \frac{9}{12} \longleftarrow \frac{3\times 3}{4\times 3} \quad \text{and} \quad \frac{5}{12}$$

$$\frac{5}{12} < \frac{8}{12} < \frac{9}{12}$$

Compare numerators

$$\therefore \frac{5}{12} < \frac{2}{3} < \frac{3}{4}$$

Order fractions by size

If comparing improper fractions, use the same method but leave the answer as a mixed numeral.

(ii) $\frac{11}{4}$ and $\frac{13}{5}$

$$\frac{11}{4} \quad , \quad \frac{13}{5}$$

LCM of denominators $= 20$

$$\frac{11\times 5}{4\times 5} \longrightarrow \frac{55}{20} \quad , \quad \frac{52}{20} \longleftarrow \frac{13\times 4}{5\times 4}$$

$$\frac{55}{20} > \frac{52}{20}$$

Compare numerators

$$\therefore 2\frac{3}{4} > 2\frac{3}{5}$$

Write in simplest form

Comparing fractions

1 Compare the size of these fractions:

a $\frac{2}{5}$ and $\frac{1}{3}$

b $\frac{3}{4}$ and $\frac{5}{7}$

2 Compare the size of the fractions in each of these groups:

a $\frac{3}{5}$, $\frac{1}{2}$ and $\frac{11}{20}$

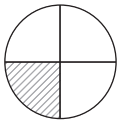
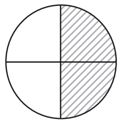
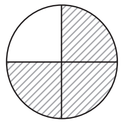
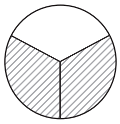
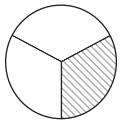
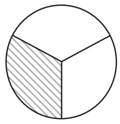
b $\frac{9}{12}$, $\frac{2}{3}$ and $\frac{5}{6}$

3 Compare the size of these improper fractions:

a $\frac{9}{4}$ and $\frac{16}{7}$

b $\frac{14}{3}$, $\frac{15}{4}$ and $\frac{21}{8}$

Adding and subtracting fractions with the same denominator

	+		=	
$\frac{1}{4}$	$+$	$\frac{2}{4}$	$=$	$\frac{3}{4}$
one quarter	and	two quarters	equals	three quarters
	–		=	
$\frac{2}{3}$	$-$	$\frac{1}{3}$	$=$	$\frac{1}{3}$
two thirds	less	one third	equals	one third

If the denominator (bottom) is the same, just add or subtract the numerators (top).

Simplify these fractions with the same denominator

(i) $\frac{2}{9}+\frac{5}{9}$

$$\frac{2}{9}+\frac{5}{9} = \frac{2+5}{9}$$ Add the numerators **only**

$$= \frac{7}{9}$$

(ii) $\frac{6}{7}-\frac{2}{7}$

$$\frac{6}{7}-\frac{2}{7} = \frac{6-2}{7}$$ Subtract the numerators **only**

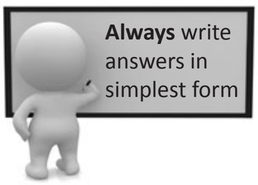
$$= \frac{4}{7}$$

(iii) $\frac{2}{3}+\frac{5}{3}$

$$\frac{2}{3}+\frac{5}{3} = \frac{2+5}{3}$$ Add the numerators **only**

$$= \frac{7}{3}$$

$$= 2\frac{1}{3}$$ Simplify



(iv) $\frac{3}{5}-\frac{1}{5}+\frac{4}{5}$

$$\frac{3}{5}-\frac{1}{5}+\frac{4}{5} = \frac{3-1+4}{5}$$ Subtract/add the numerators **only**

$$= \frac{6}{5}$$

$$= 1\frac{1}{5}$$ Simplify

Adding and subtracting fractions with the same denominator

1 Simplify these without the aid of a calculator:

a $\frac{1}{3}+\frac{1}{3}$

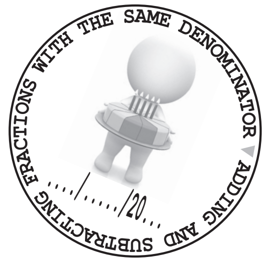
b $\frac{3}{5}-\frac{1}{5}$

c $\frac{5}{9}+\frac{2}{9}$

d $\frac{8}{11}-\frac{6}{11}$

e $\frac{11}{15}-\frac{4}{15}$

f $\frac{3}{8}+\frac{5}{8}$



2 Simplify these without the aid of a calculator:

a $\frac{1}{2}+\frac{4}{2}$

b $\frac{8}{5}-\frac{2}{5}$

c $\frac{2}{3}+\frac{5}{3}$

d $\frac{10}{4}-\frac{1}{4}$

e $\frac{11}{7}+\frac{4}{7}$

f $\frac{15}{2}-\frac{8}{2}$

3 Simplify these without the aid of a calculator, remembering to write the answer in simplest form:

a $\frac{11}{4}-\frac{5}{4}$

b $\frac{13}{6}+\frac{19}{6}$

c $\frac{9}{8}+\frac{13}{8}$

4 Simplify these without the aid of a calculator:

a $\frac{4}{9}+\frac{1}{9}+\frac{2}{9}$

b $\frac{20}{3}-\frac{10}{3}-\frac{4}{3}$

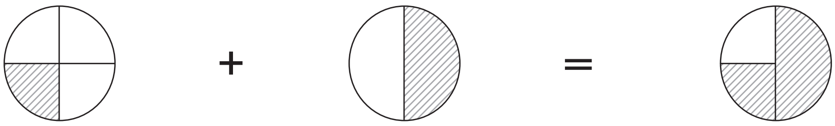
c $\frac{1}{2}+\frac{1}{2}-\frac{1}{2}$

d $\frac{1}{5}+\frac{4}{5}-\frac{2}{5}$

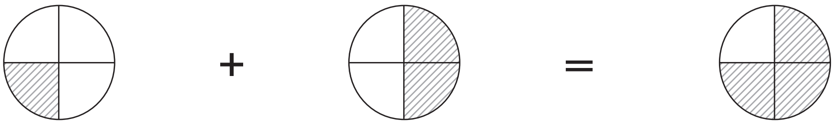
e $\frac{8}{7}-\frac{4}{7}+\frac{6}{7}$

f $\frac{13}{6}+\frac{11}{6}-\frac{9}{6}$

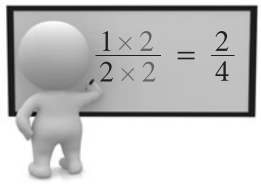
Adding and subtracting fractions with a different denominator



$\frac{1}{4}$	$+$	$\frac{1}{2}$	$=$	?
one quarter	and	one half	equals	?



$\frac{1}{4}$	$+$	$\frac{2}{4}$	$=$	$\frac{3}{4}$
one quarter	and	two quarters	equals	three quarters



Simplify these expressions which have fractions with different denominators

(i) $\frac{2}{3}+\frac{1}{5}$

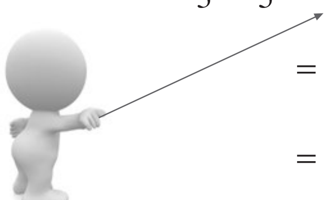
For $\frac{2}{3}$ and $\frac{1}{5}$ ← Denominators are different

$\therefore \frac{2}{3}+\frac{1}{5} = \frac{2\times 5}{3\times 5}+\frac{1\times 3}{5\times 3}$ The LCM of the denominators is 15

$= \frac{10}{15}+\frac{3}{15}$ Equivalent fractions with LCM denominators

$= \frac{10+3}{15}$ Add the numerators **only**

$= \frac{13}{15}$



Multiply top **and** bottom by the number used to make the denominator equal to the LCM

(ii) $\frac{7}{8}-\frac{1}{2}+\frac{3}{4}$

For $\frac{7}{8}$, $\frac{1}{2}$ and $\frac{3}{4}$ ← Denominators are all different

$\frac{7}{8}-\frac{1}{2}+\frac{3}{4} = \frac{7}{8}-\frac{1\times 4}{2\times 4}+\frac{3\times 2}{4\times 2}$ The LCM of all the denominators is 8

$= \frac{7}{8}-\frac{4}{8}+\frac{6}{8}$ Equivalent fractions with LCM in the denominators

$= \frac{7-4+6}{8}$

$= \frac{9}{8}$ Simplify the numerator

$= 1\frac{1}{8}$ Simplify to mixed numeral

Adding and subtracting fractions with a different denominator

1 Fill in the spaces for these calculations:

a $\frac{1}{3}+\frac{1}{6}$ The LCM of the denominators is: $\square$

$$\therefore \frac{1}{3}+\frac{1}{6} = \frac{1\times\square}{3\times\square}+\frac{1}{6}$$

$$= \frac{\square}{\square}+\frac{1}{6}$$

$$= \frac{\square}{\square}$$

$$= \frac{\square}{\square} \longleftarrow \text{simplest form}$$

b $\frac{4}{7}-\frac{1}{5}$ The LCM of the denominators is: $\square$

$$\therefore \frac{5}{7}-\frac{1}{5} = \frac{5\times\square}{7\times\square}-\frac{1\times 7}{5\times 7}$$

$$= \frac{\square}{\square}-\frac{\square}{\square}$$

$$= \frac{\square}{\square} \longleftarrow \text{simplest form}$$

2 Simplify these without the aid of a calculator:

a $\frac{1}{3}+\frac{1}{2}$

b $\frac{5}{6}-\frac{1}{2}$

c $\frac{2}{5}-\frac{1}{4}$

d $\frac{1}{6}+\frac{3}{4}$

e $\frac{6}{7}-\frac{2}{3}$

f $\frac{3}{5}+\frac{3}{8}$

Adding and subtracting fractions with a different denominator

3 Simplify these expressions without the aid of a calculator, remembering to write the answer in simplest form.

a $\frac{1}{2}+\frac{4}{5}$

b $\frac{13}{8}-\frac{3}{5}$

c $\frac{1}{2}+\frac{3}{8}-\frac{1}{4}$

d $\frac{3}{5}+\frac{3}{10}-\frac{3}{4}$

e $\frac{2}{3}-\frac{1}{4}+\frac{5}{6}$

f $\frac{7}{12}-\frac{1}{3}+\frac{11}{24}$

Adding and subtracting fractions with a different denominator

The same rules apply for questions with a mix of whole numbers and fractions. Here are some examples:

Simplify these expressions which have a mix of whole numbers and fractions

(i) $3+\frac{1}{4}$

$3+\frac{1}{4} = 3\frac{1}{4}$ — Write the fraction after the whole number

(ii) $1-\frac{2}{5}$

$1-\frac{2}{5} = \frac{5}{5}-\frac{2}{5}$ — Write the whole number as a fraction with same denominator

$= \frac{3}{5}$ — Subtract the numerators **only**

(iii) $4-\frac{2}{7}$

$4-\frac{2}{7} = \frac{28}{7}-\frac{2}{7}$ — Write the whole number as a fraction with same denominator

$= \frac{26}{7}$

$= 3\frac{5}{7}$ — Simplify fraction

4 Simplify these expressions:

a $2+\frac{1}{2}$

b $1+\frac{3}{4}$

c $1-\frac{2}{3}$

d $1-\frac{3}{8}$

e $2-\frac{3}{5}$

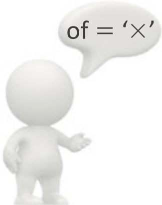
f $4-\frac{1}{4}$

g $3-\frac{5}{3}$

h $5-\frac{5}{2}$

Multiplying and dividing fractions

To **multiply** fractions, just remember: Multiply the numerators (top) and the denominators (bottom)



$$\frac{1}{3} \text{ of } \frac{2}{5} = \frac{1}{3} \times \frac{2}{5} = \frac{1 \times 2}{3 \times 5} = \frac{2}{15}$$

To **divide** an amount by a fraction, just remember: flip the second fraction then multiply

$$\frac{1}{3} \div \frac{2}{5} = \frac{1}{3} \times \frac{5}{2}$$

Only flip the second fraction

$$= \frac{1 \times 5}{3 \times 2}$$

Change the '÷' to a '×'

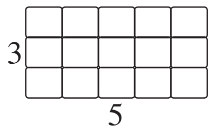
$$= \frac{5}{6}$$

Remember: A flipped fraction is called the **reciprocal** fraction

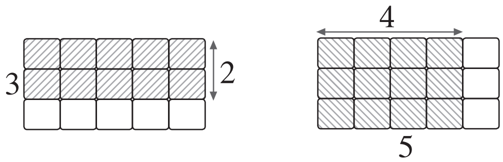
Simplify these:

We can use shaded diagrams to calculate the multiplication of two fractions

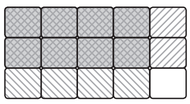
(i) $\frac{2}{3}$ of $\frac{4}{5}$



Draw a grid using the denominators as the dimensions



Use the numerators to shade columns/rows



$= \frac{8}{15}$

$$\therefore \frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$$

Write where they **overlap** as a fraction

If whole numbers are involved, write them as a fraction

(ii) $28 \div \frac{2}{7}$

$$\therefore 28 \div \frac{2}{7} = 28 \times \frac{7}{2}$$

Flip the second fraction and change sign to '×'

$$= \frac{28}{1} \times \frac{7}{2}$$

Write the whole number as a fraction

$$= \frac{196}{2}$$

$$= \frac{98}{1}$$

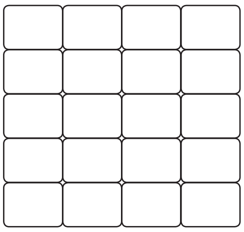
Simplify

$$= 98$$

Multiplying and dividing fractions

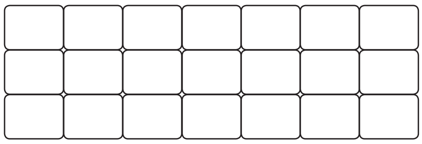
1 Calculate these fraction multiplications by shading the given grids:

a $\frac{1}{5}$ of $\frac{3}{4}$

5

4

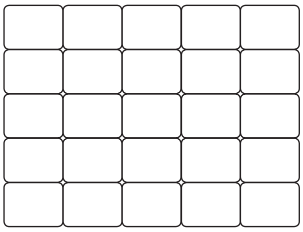
$\therefore \frac{1}{5}$ of $\frac{3}{4} = \frac{\square}{\square}$

b $\frac{2}{3}$ of $\frac{4}{7}$

3

7

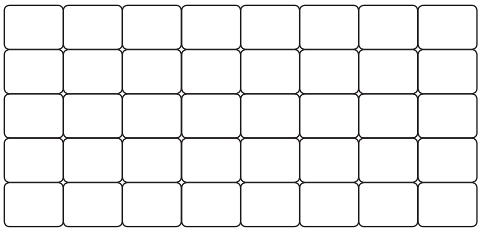
$\therefore \frac{2}{3}$ of $\frac{4}{7} = \frac{\square}{\square}$

c $\frac{4}{5}$ of $\frac{4}{5}$

5

5

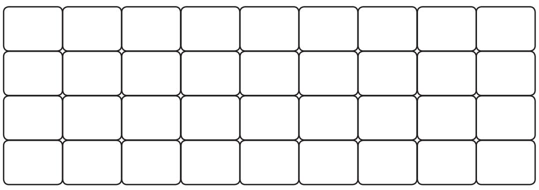
$\therefore \frac{4}{5}$ of $\frac{4}{5} = \frac{\square}{\square}$

d $\frac{2}{5}$ of $\frac{3}{8}$

5

8

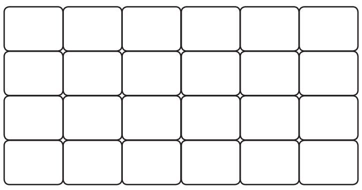
$\therefore \frac{2}{5}$ of $\frac{3}{8} = \frac{\square}{\square} = \frac{\square}{\square}$ simplified

e $\frac{3}{4}$ of $\frac{7}{9}$

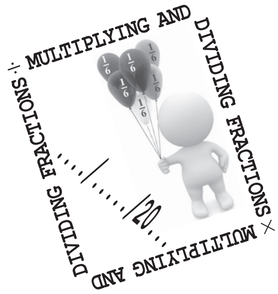
4

9

$\therefore \frac{3}{4}$ of $\frac{7}{9} = \frac{\square}{\square} = \frac{\square}{\square}$ simplified

f $\frac{3}{4}$ of $\frac{5}{6}$

4

6

$\therefore \frac{3}{4}$ of $\frac{5}{6} = \frac{\square}{\square} = \frac{\square}{\square}$ simplified



Multiplying and dividing fractions

2 Simplify these without the aid of a calculator:

a $\frac{1}{2} \times \frac{1}{3}$

b $\frac{3}{5} \times \frac{1}{4}$

c $\left(\frac{2}{3}\right)^2$ psst: this is just $\frac{2}{3} \times \frac{2}{3}$

d $\left(\frac{3}{5}\right)^2$

e $\frac{1}{3} \div \frac{3}{2}$

f $\frac{2}{11} \div \frac{1}{4}$

g $\frac{5}{6} \div 4$

h $\frac{3}{4} \div 8$

i $10 \times \frac{4}{5}$

j $24 \times \frac{3}{8}$

k $12 \div \frac{3}{5}$

l $2 \div \frac{2}{13}$

Multiplying and dividing fractions

3 Simplify these without the aid of a calculator, remembering to write the answer in simplest form.

a $\left(\frac{2}{8}\right)^2$

b $\frac{3}{4} \times \frac{3}{2}$

c $\frac{3}{8} \div \frac{5}{4}$

d $\frac{2}{3} \div \frac{5}{3}$

e $\frac{9}{10} \div \frac{8}{5}$

f $\frac{3}{4} \times \frac{2}{3} \times \frac{1}{2}$ psst: same as the others!

g $\frac{2}{5} \times \frac{3}{6} \times \frac{1}{3}$

h $\frac{1}{2} \div 4 \div \frac{1}{2}$ psst: work left to right!

4 Is $\frac{2}{3}$ of $\frac{4}{6}$ exactly the same as $\frac{2}{3} \div \frac{12}{8}$? Explain your answer.

Operations with mixed numerals

Just change to improper fractions then use the same methods as shown earlier.

Simplify these calculations involving mixed numerals

Addition and subtraction

(i) $1\frac{2}{3} + 2\frac{1}{6}$

$$1\frac{2}{3} + 2\frac{1}{6} = \frac{5}{3} + \frac{13}{6}$$ Change to improper fractions

$$= \frac{10}{6} + \frac{13}{6}$$ Equivalent fractions with LCM denominators

$$= \frac{23}{6}$$

$$= 3\frac{5}{6}$$ Simplify to mixed numeral

Or just add the whole numbers and the fractions separately.

$1 + 2 = 3 \quad \frac{2}{3} + \frac{1}{6} = \frac{5}{6}$

(ii) $4\frac{1}{5} - 1\frac{1}{2}$

$$4\frac{1}{5} - 1\frac{1}{2} = \frac{21}{5} - \frac{3}{2}$$ Change to improper fractions

$$= \frac{42}{10} - \frac{15}{10}$$ Equivalent fractions with LCM denominators

$$= \frac{27}{10}$$

$$= 2\frac{7}{10}$$ Simplify to mixed numeral

Multiplication and division

(iii) $1\frac{3}{4} \times 2\frac{1}{3}$

$$1\frac{3}{4} \times 2\frac{1}{3} = \frac{7}{4} \times \frac{7}{3}$$ Change to improper fractions

$$= \frac{49}{12}$$ Multiply tops and bottoms together

$$= 4\frac{1}{12}$$ Simplify to mixed numeral

(iv) $1\frac{1}{6} \div 2$

$$1\frac{1}{6} \div 2 = \frac{7}{6} \div \frac{2}{1}$$ Change to improper fractions

$$= \frac{7}{6} \times \frac{1}{2}$$ Flip second fraction and change to multiply

$$= \frac{7}{12}$$ Multiply numerators and denominators together

Remember

$2 = \frac{2}{1}, 3 = \frac{3}{1}$, etc

Operations with mixed numerals

1 Simplify these additions and subtractions without the aid of a calculator:

a $2\frac{1}{4}+4\frac{2}{3}$

b $2\frac{1}{4}-1\frac{2}{5}$

c $5\frac{3}{5}-2\frac{1}{2}$

d $3\frac{1}{6}+1\frac{1}{5}$

2 Simplify these without the aid of a calculator:

a $4\times1\frac{2}{5}$

b $4\frac{3}{7}\times2$

c $1\frac{3}{4}\times3\frac{1}{2}$

d $5\frac{1}{3}\times1\frac{4}{5}$

3 Simplify these divisions without the aid of a calculator:

a $3\div2\frac{1}{2}$

b $1\frac{2}{3}\div3$

c $2\frac{2}{5}\div1\frac{1}{2}$

d $5\frac{1}{2}\div1\frac{2}{3}$

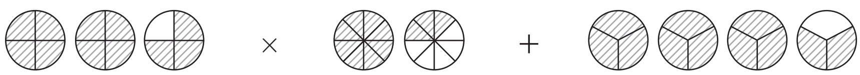
Combining all the operations

Earn yourself an awesome passport stamp by trying these trickier questions without using a calculator.

1. Simplify $\frac{1}{7}+6\frac{3}{5}\times 1\frac{1}{2}$ psst: remember your order of operations

2. Simplify $4\frac{1}{2}\div 3\frac{5}{6}\times 5\frac{1}{3}$ psst: work left to right… so do the division first

3. Simplify this shaded diagram into a single fraction. psst: write as fractions, then work left to right



Fractions of an amount

How many links are there in $\frac{2}{5}$ of a chain made using a total of 20 links?

Simplified, this question is just: Find $\frac{2}{5}$ of 20.

$$\begin{aligned}\therefore \frac{2}{5} \text{ of } 20 &= \frac{2}{5}\times 20\\ &= \frac{2}{5}\times\frac{20}{1}\\ &= \frac{40}{5}\\ &= 8\end{aligned}$$

$\therefore \frac{2}{5}$ of the 20 links $= 8$ links

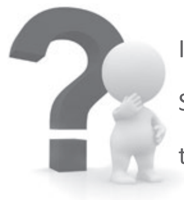
Remember: ‘of’ = ‘×’

Here are some questions that calculate fractions of an amount.

(i) Juliet lost $\frac{1}{4}$ of the 116 songs she had downloaded when a computer virus infected the files.
How many songs were **not** affected by the virus?

$$\begin{aligned}\frac{1}{4} \text{ of } 116 \text{ songs} &= \frac{1}{4}\times 116\\ &= \frac{1}{4}+\frac{116}{1}\\ &= \frac{116}{4}\\ &= 29\end{aligned}$$

$\therefore 116 - 29 = 87$ songs not affected by the virus — Answer the question



If $\frac{1}{4}$ are gone, then $\frac{3}{4}$ remain.
So finding $\frac{3}{4}$ of 116 will get the same answer. Try it!

(ii) How long is $\frac{7}{10}$ of 1 hour?

$$\begin{aligned}\frac{7}{10} \text{ of } 1 \text{ hour} &= \frac{7}{10} \text{ of } 60 \text{ minutes}\\ &= \frac{7}{10}\times\frac{60}{1}\\ &= \frac{420}{10}\\ &= 42\end{aligned}$$

Change to smaller units if possible

$\therefore \frac{7}{10}$ of 1 hour $= 42$ minutes — Answer the question

Fractions of an amount

1 Calculate the amount for each of these, showing all working:

a $\frac{1}{5}$ of 20

b $\frac{3}{4}$ of 32

c $\frac{2}{3}$ of 24

d $\frac{5}{6}$ of 42

2 Calculate the amount for each of these by first making the mixed numeral an improper fraction:
psst: the answers will be bigger than the given whole number

a $2\frac{1}{2}$ of 4

b $1\frac{4}{5}$ of 15

c $3\frac{2}{7}$ of 14

d $4\frac{2}{3}$ of 36

3 Calculate these fractions of quantities, showing all working:

a How many hours is $\frac{3}{4}$ of 1 day?

1 day $=$ 24 hours

b How many grams is $\frac{3}{10}$ of 2 kilograms?

1 kg $=$ 1000 grams

Fractions of an amount

3 c How long is $\frac{5}{6}$ of 2 hours?

1 hour $=$ 60 minutes

d How far is $\frac{1}{5}$ of 3 kilometres?

1 km $=$ 1000 metres

4 In an orchestra of 60 musicians, $\frac{1}{5}$ were in the brass section. How many brass section players were there?

psst: remember to include a statement answering the question at the end

5 Krista and her team mates each receive $\frac{1}{5}$ of a $\$900$ prize for winning a competition.
How much does Krista receive?

6 Hank bought $\frac{2}{7}$ of the 28 towels that were on sale in a shop. How many towels were **not** bought by Hank?

7 The lead in one brand of HB pencil is $\frac{8}{11}$ graphite.

How many grams of non-graphite material are there in this brand if every pencil contains 33 grams of lead?

Fractions of an amount

Here is what to do if you already have the fraction amount and need to find the original whole amount.

Amani used 250 g of one ingredient, which was $\frac{2}{5}$ of the total ingredients used in the recipe she was following. What is the total weight of ingredients used in this recipe?

A short-cut way is to multiply 250 g by the reciprocal.

$250\text{ g} \times \frac{5}{2} = 625\text{ g}$

$$\begin{aligned}
\tfrac{2}{5}\text{ of the total ingredients used} &= 250\text{ g} && \text{Divide the amount by the numerator}\\
\therefore \tfrac{1}{5}\text{ of the total ingredients} &= 250\text{ g} \div 2 \\
&= 125\text{ grams} && \text{Multiply answer by the denominator}\\
\therefore \tfrac{5}{5}\text{ (the total ingredients)} &= 125\text{ g} \times 5 \\
&= 625\text{ grams}
\end{aligned}$$

$\therefore$ Total weight of ingredients used in the recipe $= 625$ grams

8. The letter ‘e’ is used twenty one times in this question. If this is $\frac{1}{8}$ of all the letters used to write this question, work out how many letters there are in total (show all working and check your amount by counting).

9. During a performance by Jolly Rob, 280 members of the audience found his jokes funny. If this was $\frac{4}{7}$ of the total audience members at the performance, how many people were watching Jolly Rob?

10. By $3{:}00$ pm, Juan had been waiting 30 minutes for his bus to arrive. If this is $\frac{5}{6}$ of the total time he spent waiting, at what time did his bus arrive?

Two amounts as a fraction

Sometimes we need to compare two amounts by writing one as a fraction of the other.
For example, write 14 out of 36 as a fraction in simplest form

First amount → 14; Second 'out of' amount → 36

$$\frac{14}{36} = \frac{14 \div 2}{36 \div 2}$$ Divide the amount by the HCF

$$= \frac{7}{18}$$ Simplest form

$$\therefore 14 \text{ out of } 36 = \frac{7}{18}$$

These examples show how both amounts must be in the same units before writing as a fraction.

Write these amounts as fractions of one another in simplest form

(i) All the boxes in the picture weigh a total of 40 kg. The box being carried is 2000 g.
What fraction of the total weight is being carried?

$1 \text{ kg} = 1000 \text{ g}$

$40 \text{ kg} = 40\,000 \text{ g}$ Need both weights in the same **smaller** units

$\therefore \dfrac{2000}{40\,000}$ = the fraction of the total weight being carried

$= \dfrac{1}{20}$ Simplify fraction

$\therefore$ the box being carried is $\dfrac{1}{20}$ of the total weight Answer question

(ii) What fraction of $2\frac{1}{2}$ hours is 15 minutes?

$2\frac{1}{2}$ hours $= 2\frac{1}{2} \times 60$ minutes $1 \text{ hour} = 60$ minutes

$= 150$ minutes Need both times in the same **smaller** units

First smaller amount → 15; total amount → 150

$\dfrac{15}{150} = \dfrac{1}{10}$ Simplify fraction

$\therefore 15$ minutes is $\dfrac{1}{10}$ of $2\frac{1}{2}$ hours Answer question

(iii) If one ream of paper contains 500 sheets, what fraction is 240 sheets out of 4 reams?

4 reams $= 4 \times 500$ sheets

$= 2000$ sheets Need both amounts in the same **smaller** units

First smaller amount → 240; total amount → 2000

$\dfrac{240}{2000} = \dfrac{3}{25}$ Simplify

$\therefore 240$ sheets is $\dfrac{3}{25}$ of 4 reams Answer question

Two amounts as a fraction

1. Write the first amount as a fraction of the second for each of these in simplest form

a. 5 out of 20

b. 16 out of 22

c. 35 marks out of a possible 40

d. 2 hours of 1 day (1 day = 24 hours)

e. 5 minutes of half-an-hour (1 hour = 60 minutes)

f. 25 cents out of \$2 (\$1 = 100 cents)

g. 300 seconds of 1 hour (1 hour = 3600 seconds)

h. 23 days of March (31), April (30) and May (31)

2. In a school of 800 students, 240 were in Year 7. What fraction of the school are Year 7 students?

3. After 100 test rolls, a die displayed the number 5 sixteen times. What fraction of the rolls were 5?

4. Francesca fired 27 arrows at a target and hit the bullseye 6 times. What fraction of arrows **missed** the bullseye?

5. A biscuit recipe contained 500 g of flour, 175 g of sugar and 125 g of butter. What fraction of the recipe is butter?

Word problems with fractions

While on a shopping trip, Xieng spent two fifths of her money on clothes and one third on cosmetics. What fraction of her money did Xieng have left?

$$\begin{aligned}\frac{2}{5}+\frac{1}{3} &= \text{fraction of Xieng's money spent on shopping}\\ &= \frac{6+5}{15}\\ &= \frac{11}{15} \qquad \text{Add the numerators together}\\ \therefore \frac{15}{15}-\frac{11}{15} &= \frac{4}{15}\end{aligned}$$

Fraction for all of Xieng's money   Fraction spent   Fraction of money Xieng has left

$\therefore$ Xieng still has $\frac{4}{15}$ of her money after shopping

Here are some other word problem examples

(i) In a group of eighteen friends, one third are girls and one sixth of these girls have blonde hair. How many blonde girls are in the group?

$$\begin{aligned}\therefore \frac{1}{6} \text{ of } \frac{1}{3} \text{ of } 18 &= \text{number of blonde girls in the group}\\ &= \frac{1}{6}\times\frac{1}{3}\times\frac{18}{1}\\ &= \frac{18}{18}\\ &= 1\end{aligned}$$

$\therefore$ There is 1 blonde girl in the group of friends.

(ii) During one night, possums ate two fifths of the fifty five fruits on a tree. If one eleventh of the eaten fruit grew back, how many fruits are now on the tree?

$$\begin{aligned}\frac{2}{5}\times 55 &= \text{Number of fruits eaten}\\ &= \frac{110}{5}\\ &= 22\\ \frac{1}{11}\times 22 &= \text{Number of fruits that grew back}\\ &= \frac{22}{11}\\ &= 2\\ \therefore \text{Number of fruits now on the tree} &= 55-22+2\\ &= 35 \text{ pieces of fruit}\end{aligned}$$

Word problems with fractions

1. At a recent trivia night, one table of competitors answered five eighths of the fifty six questions correctly. How many questions did they get **incorrect**?

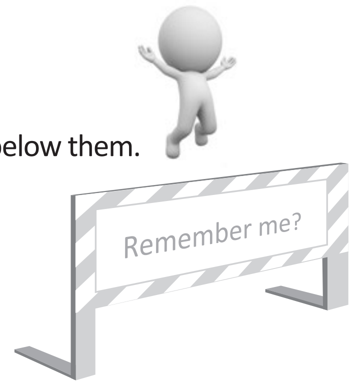
2. Co Tin usually takes approximately sixty and one quarter steps every minute when walking. How many steps does he expect to take when he exercises by walking for one and two third hours each day?

3. A vegetable garden has one third carrots, one sixth pumpkins, one quarter herbs, and the rest are potato plants. How many potato plants are in this garden of eighty plants?

4. A class of twenty four students compared eye colours on a chart. Two thirds of the class had brown eyes, and three eighths of those brown-eyed students were boys. How many girls had brown eyes?

Word problems with fractions

5 For one particular school:
There are 256 students in Year 7.
The Year 8, 9 and 10 groups all have half the number of students than the year just below them.
How many students are there at this school in Years 7 to 10?



6 Five students in a class have a combined total of ninety profiles added as friends on a web-based social network site.
Three fifths of the ninety profiles are shared by all five of the students. These shared profiles represent one sixth of the total number of different profiles added as friends by all the students in the class.
How many different profiles are linked to students from this class?

7 Five sevenths of the fifty six images used as backgrounds on Meagan's touchpad were photos she took herself. After moving five eighths of these photos to another computer, what fraction of the background images now are **not** photos taken by her?

Reflection Time

Reflecting on the work covered within this booklet:

1. What useful skills have you gained by learning about fractions?

2. Write about one way you think you could apply fractions to a real life situations.

3. If you discovered or learnt about any shortcuts to help with fractions or some other cool facts, jot them down here:

Here is a summary of the things you need to remember for fractions

Proper fractions

Represent parts of a whole number or object. The numerator is smaller than or equal to the denominator.

numerator $\longrightarrow$ $\dfrac{1}{2}$ $\longleftarrow$ number of equal parts **you have**
denominator $\longrightarrow$ $\longleftarrow$ **total** number of equal parts

Equivalent proper fractions

These are fractions with different numbers that represent the same amount.

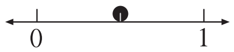
$$\therefore \frac{4}{8} = \frac{2}{4} = \frac{1}{2} = \text{Equivalent fractions}$$

Improper fractions and mixed numerals

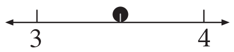
$\frac{3}{2} \longleftarrow$ Improper fractions $\longrightarrow \frac{5}{4}$ numerator > denominator

$1\frac{1}{2} \longleftarrow$ Mixed numerals $\longrightarrow 1\frac{1}{4}$ A 'mix' of whole numbers and proper fractions.

Fractions on the number line



$\dfrac{1}{2}$ $\longleftarrow$ number of equal steps taken between 0 and 1
$\longleftarrow$ **total** number of **equal** steps between 0 and 1



Start $\nearrow$ $3\dfrac{1}{2}$ $\longleftarrow$ number of equal steps towards the next whole number
$\longleftarrow$ **total** number of equal steps between start and next whole number

Reciprocal fractions

Original fraction $\longrightarrow \frac{2}{5} \longrightarrow \frac{5}{2} \longleftarrow$ Reciprocal fraction

Mixed numeral $\longrightarrow 3\frac{1}{2} \longrightarrow \frac{7}{2} \longrightarrow \frac{7}{2} \longrightarrow \frac{2}{7} \longleftarrow$ Reciprocal fraction

Comparing fractions

Write equivalent fractions by changing the denominators to their LCM, then compare the numerators

$$\frac{1\times 3}{2\times 3} \longrightarrow \frac{3}{6} > \frac{2}{6} \longleftarrow \frac{1\times 2}{3\times 2}$$

Adding and subtracting fractions

If the denominators (bottom) are the same, then simply add or subtract the numerators (tops).
If the denominators are the different, change to equivalent fractions with the same denominators using the LCM. Then add or subtract the numerators of the new fractions.

Multiplying and dividing fractions

To **multiply** fractions, just remember: Multiply the numerators (top) and the denominators (bottom).
To **divide** an amount by a fraction, just remember: flip the second fraction (reciprocal) then multiply.

Fractions of an amount

'of' means '$\times$' $\therefore$ Find $\frac{2}{5}$ of 2 means calculate $\frac{2}{5} \times 20$

Two amounts as a fraction

2 out of 5 as a fraction is $\frac{2}{5}$. If the two amounts are in different units, change the larger amount into the smaller units. Eg, 200 g out of 2 kg becomes 200 g out of 2000 g.